

Development of an Application for Supporting Online Purchase Decisions

Gkovaris Christos-Grigorios

University of Ioannina

Department of Computer Science and Engineering

Supervisor: *Anastasiadis Stergios*

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MOTIVATION

The Problem

List without Priorities

Saving products does not indicate which purchase should come first

Multiple Criteria

Urgency, discounts, prices, and user preferences must be considered

Limited Budget

Not every product or combination is financially feasible

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GOAL

Objective of the Project

Browser-Based Application

Development of a **Chrome Extension** that supports online purchase decisions using **deterministic**, **rule-based methods**

- Import real product data from Skroutz Favorites
- Prioritize products based on urgency, discounts, and budget constraints
- Generate explainable, budget-aware purchase plans

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THEORY

Theoretical Background

Decision Support

Helping users compare products and make informed purchasing decisions

Multi-Criteria Evaluation

Evaluating products using explicit criteria and user-defined preference weights

Budget Constraint

Selecting feasible product combinations without exceeding the available budget

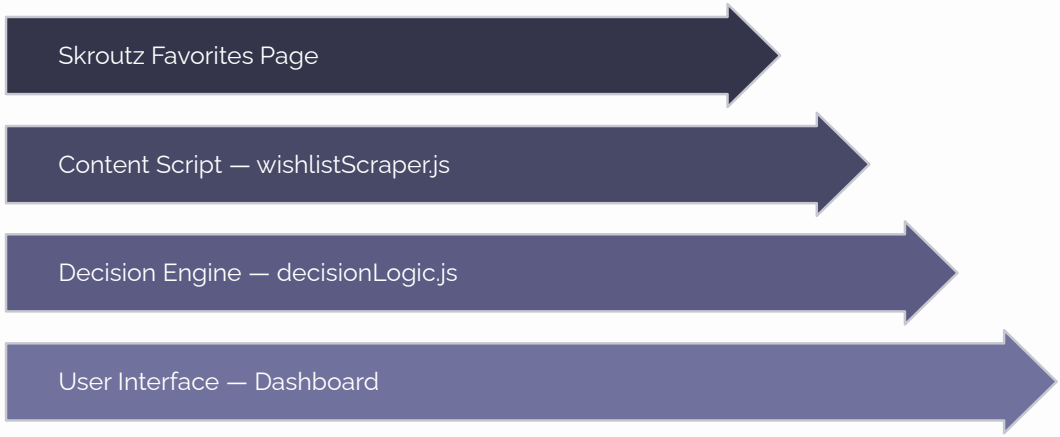
Explainability

Providing understandable scores and reasons behind purchase recommendations

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DESIGN

System Architecture



Tech Stack

- HTML5 · CSS3 · JavaScript
- Chrome Extensions Manifest V3
- DOM API
- Chrome Extension APIs
- Chrome Storage API

Data Flow

DOM Extraction → Chrome Messaging → Decision Processing → Results Display

Local-first architecture

No backend or external database

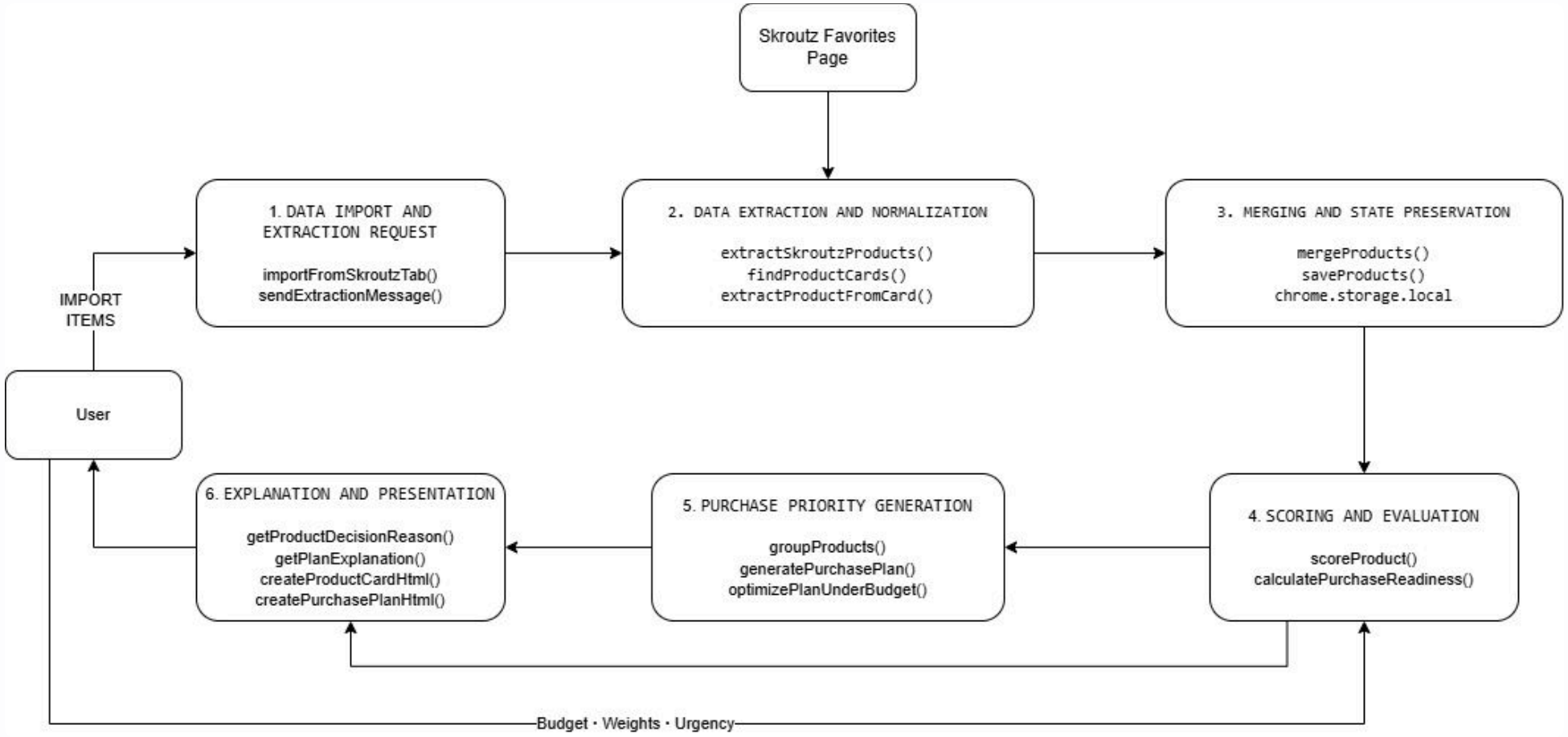
Chrome Local Storage

Persistent products, budgets, preferences and lists

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DESIGN

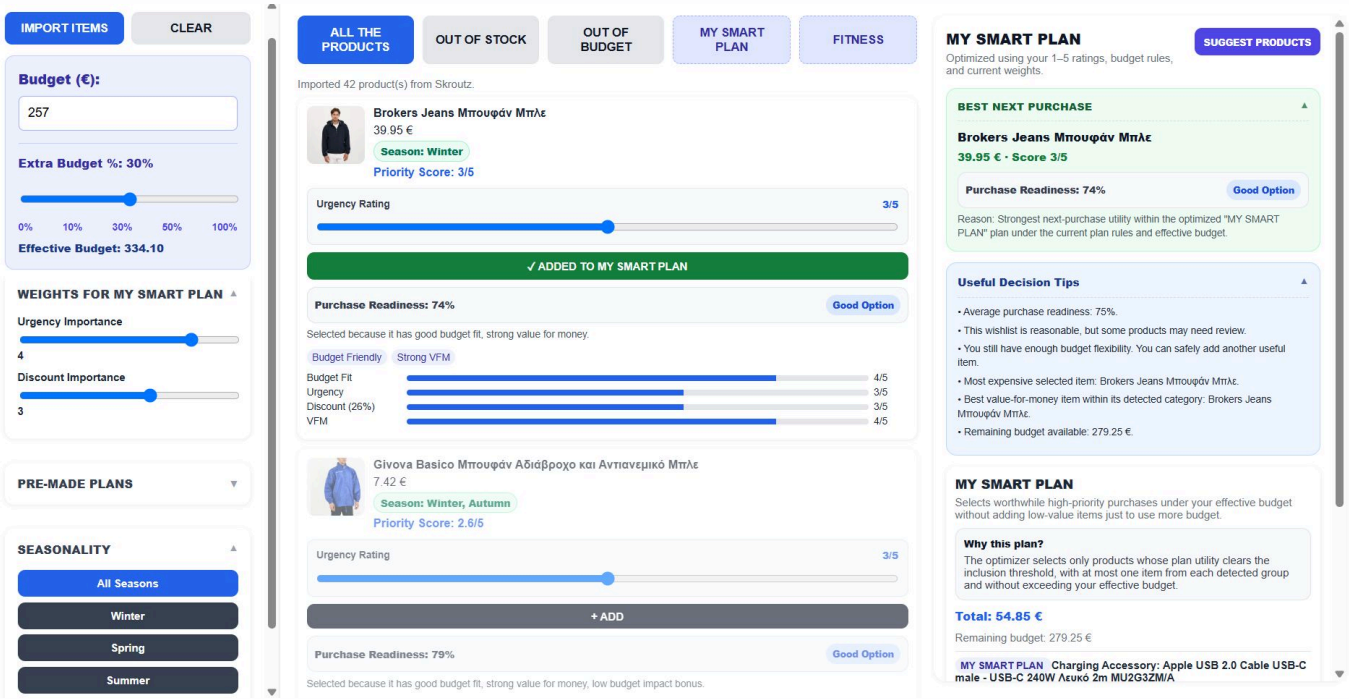
Application Structure Diagram



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INTERFACE

User Interface



Settings

Budget, preference weights, purchase strategies and seasonality filter

Product Evaluation

Imported products, urgency ratings, Priority Scores and Purchase Readiness

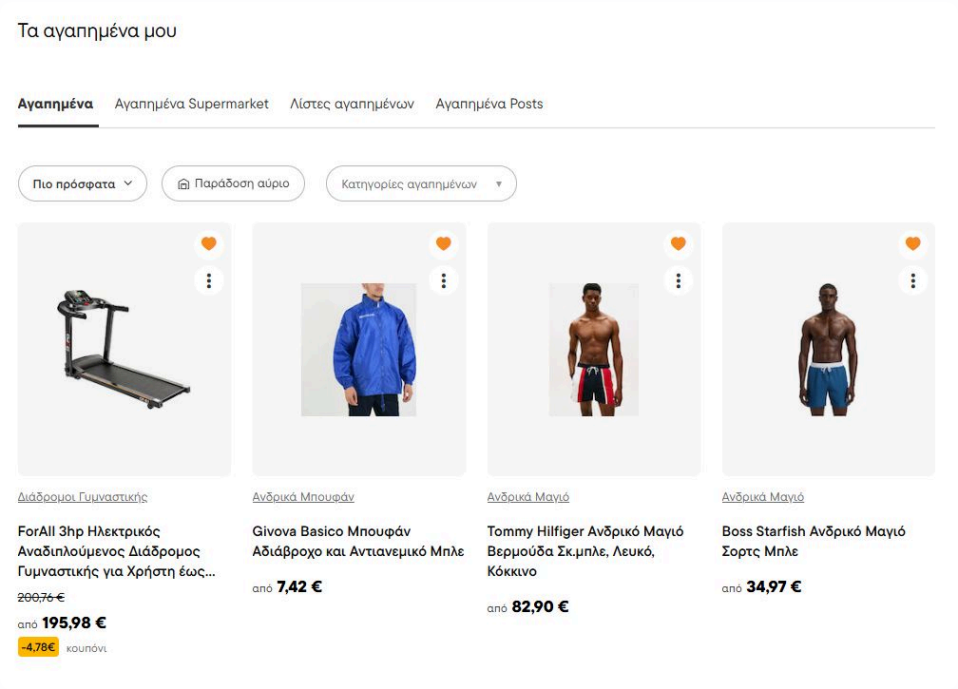
Purchase Planning

Optimized purchase plan, Best Next Purchase and decision explanations

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DATA COLLECTION

Product Import from Skroutz



1

Open Favorites

The user opens their Skroutz Favorites page

2

Start Import

The user clicks IMPORT ITEMS in the extension

3

Extract Product Data

The content script reads available product fields from the page DOM

4

Transfer to Application

The extracted product records are transferred to the application for processing

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DATA PIPELINE

Data Processing & Persistence

Price Normalization

Parse Greek-formatted euro prices, previous prices and coupon discounts into consistent numeric values

Availability Validation

Detect explicitly unavailable products. A missing price alone does not mean a product is out of stock

Deduplication & Merge

Match products using normalized URLs or titles, while preserving user-defined settings during re-import

Local Persistence

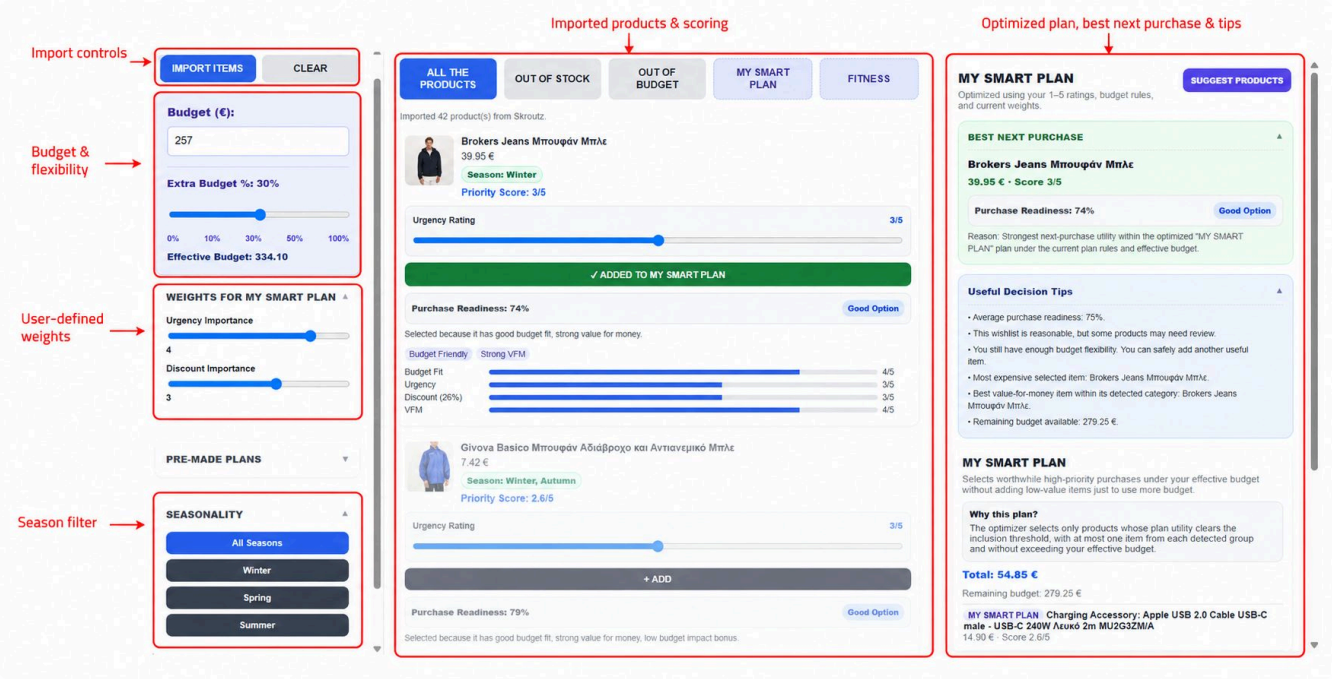
Store products, budgets, urgency ratings, tracked items, lists and preferences using chrome.storage.local

Raw Product Data → Validation & Normalization → Merge → Local Storage

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USER CONFIGURATION

Budget & User Controls



Effective Budget

Base Budget × (1 + Extra Budget % / 100)

Example: €257 + 30% = €334.10

Priority Weights

Adjustable Urgency and Discount Importance (0–5)

Purchase Strategies

My Smart Plan or predefined planning strategies

Seasonality Filter

Filters products by season; not a Priority Score weight

The effective budget is a hard constraint for purchase-plan optimization

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SCORING MODEL

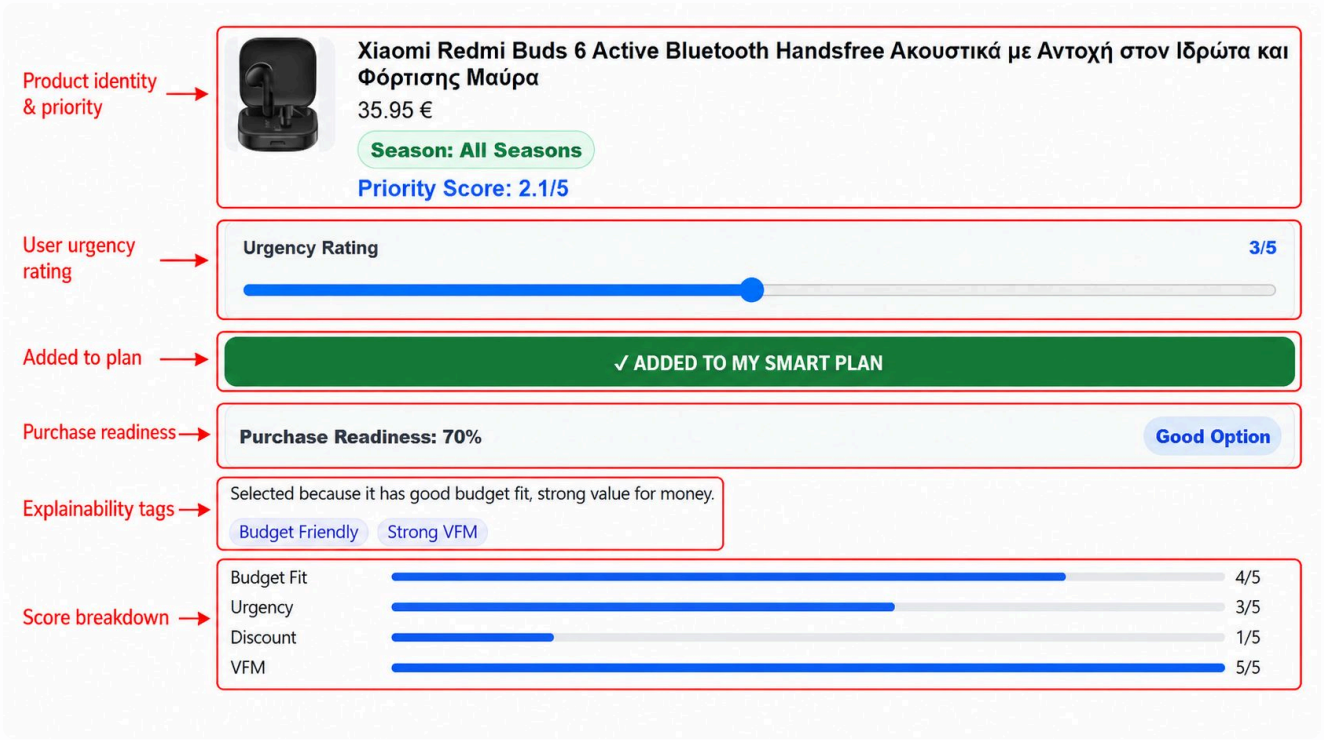
Urgency & Discount Ratings

Urgency Rating (1–5)

- Manually assigned by the user to each product
- A higher rating indicates greater personal urgency

Discount Rating (1–5)

- Calculated from detected price reductions and coupon information, when available
- A higher rating indicates a stronger detected discount



Importance Weights (0–5) determine how strongly each rating influences the Base Score

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SCORING MODEL

Priority Score Calculation

Weighted Base Score

Base Score = (U × wU + D × wD) / (wU + wD)

U = Urgency Rating (1–5) · D = Discount Rating (1–5)
wU = Urgency Importance (0–5) · wD = Discount Importance (0–5)

Calculation Example

Urgency Contribution

3 × 4 = 12

Discount Contribution

1 × 3 = 3

Total Weighted Value: 12 + 3 = 15

Total Importance Weight: 4 + 3 = 7

FINAL BASE SCORE: 15 / 7 ≈ 2.14 (Out of 5 · Before budget adjustment)

Final Priority Score

1. Calculate the weighted Base Score
2. Apply a budget-based adjustment
3. Restrict the result to the 1–5 scale

Priority Score = Clamp(Base Score + Budget Adjustment, 1, 5)

Budget Adjustment: An affordability-based bonus or penalty determined by the product price relative to the effective budget.

When both importance weights are zero, the Base Score defaults to 3

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SUPPORTING METRICS

Budget Fit & Value for Money

1

Budget Fit (1–5)

- Measures product affordability relative to the effective budget
- Higher ratings indicate a more comfortable budget fit

2

Value for Money (1–5)

- Compares product prices within their detected category
- Higher ratings indicate more favorable relative pricing

3

Role in Evaluation

- Both metrics provide additional information beyond the main Priority Score
- They are not independent weights in the Base Score formula

4

Decision Support

- Budget Fit contributes to Purchase Readiness
- Both metrics are used in purchase-planning strategies

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FILTERING

Seasonality Filtering



Purpose

Filters products according to their seasonal relevance

Available Options

All Seasons, Winter, Spring, Summer and Autumn

How It Is Used

The user selects one season to view products that match the active filter

Important Clarification

Seasonality is a filtering mechanism and it is not an additional weight in the Base Score formula

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PRODUCT VALIDATION

Product Availability & Budget Status

Available

Valid price and no explicit out-of-stock indication. Budget eligibility is checked separately

Out of Stock

The source page explicitly marks the product as unavailable (Purchase Readiness: **0%**)

Price Unavailable

Missing or invalid price data does not automatically indicate out-of-stock status (Purchase Readiness: **0%**)

Out of Budget

Product price exceeds the user's effective budget (Purchase Readiness: **10%**)

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INDICATOR

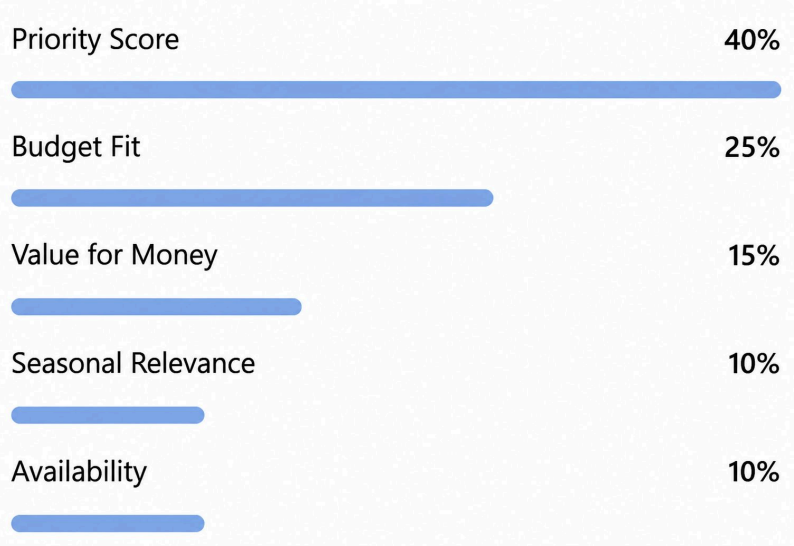
Purchase Readiness

0-100%

Weighted purchase-readiness indicator

Includes Priority Score as a weighted component · Not a purchase probability

Five Weighted Components



Components are normalized to a common percentage scale before weighting

Readiness Classification

Strong Buy **(80–100%)**

Good Option **(65–79%)**

Maybe Later **(45–64%)**

Low Priority **(Below 45%)**

Special Cases
Out of Stock / Invalid Price → 0%
Over Effective Budget → 10%

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PURCHASE PLANNING

Purchase Plan Strategies & Optimization

My Smart Plan

Uses user-defined importance weights to prioritize purchases according to personal preferences

Budget Plan

Prioritizes affordability and budget fit when selecting products

Premium Plan

Gives greater importance to relatively higher-priced products while considering their priority

Deal Hunter

Emphasizes detected discounts to identify more attractive purchase opportunities

Shared Optimization Rules

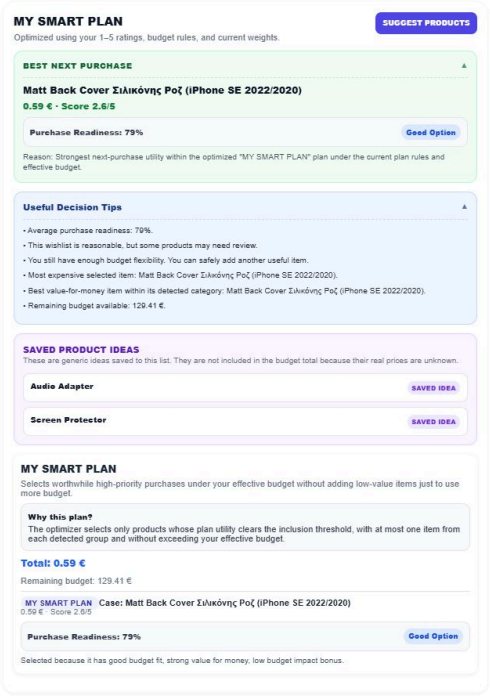
Effective Budget limit · Maximum one product per detected category · Minimum utility threshold

Select a high-utility combination without adding low-value products merely to exhaust the budget

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PURCHASE PLANNING

Optimized Plan & Best Next Purchase



Optimized Purchase Plan

A selected combination of eligible products that maximizes utility under the effective budget

Budget Summary

Displays the total plan cost and the remaining available budget

Best Next Purchase

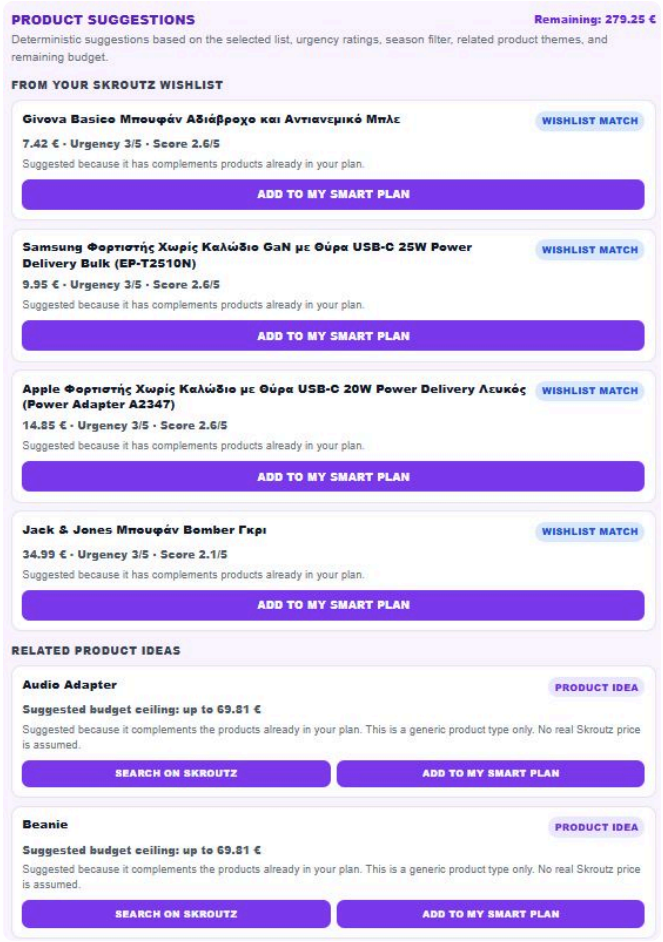
Highlights the preferred product from the generated plan according to the active strategy

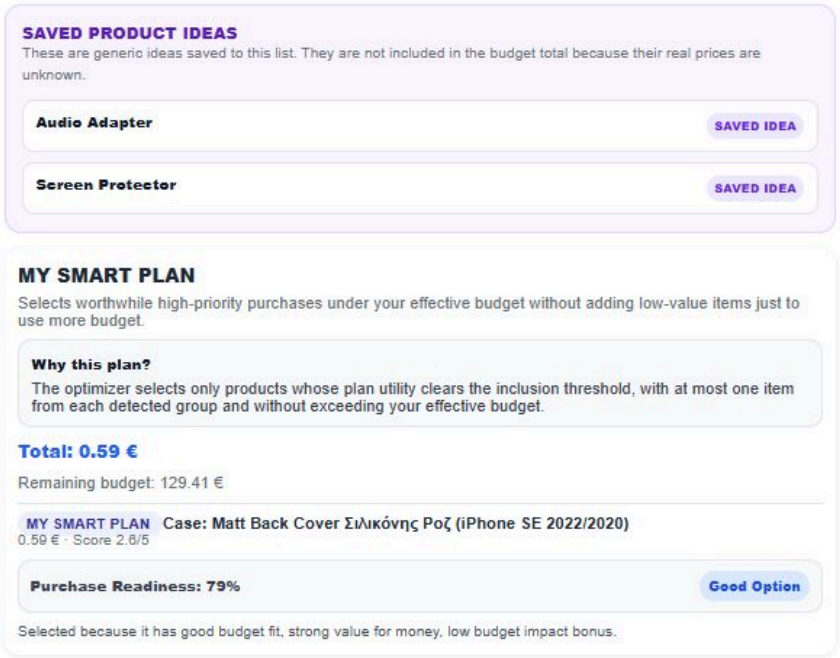
Product Evaluation → Optimized Purchase Plan → Best Next Purchase

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EXPLAINABILITY

Complementary Suggestions and Explainability





Wishlist Suggestions

Recommend relevant products from the user's imported Skroutz Favorites

Generic Product Ideas

Rule-based complementary ideas with unverified prices, excluded from budget calculations

Explainability

Decision reasons, criterion breakdowns and budget-related explanations

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TESTING

Testing & Validation

1

Automated Testing — Python unittest runner and Node.js test harness with mocked Chrome APIs

2

Functional Validation — 22 browser-based test scenarios covering importing, scoring, budgeting, and persistence

3

Edge Cases — Invalid prices, unavailable products, duplicate imports, budget boundaries, and optimizer constraints

4

Test Results — All tests passed in the recorded automated and functional validation runs

Deterministic logic · Regression testing · Boundary-case validation

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CONCLUSION

Conclusions and Future Work

Main Results

A fully client-side Chrome Extension that provides explainable product prioritization and budget-aware purchase plans using real Skroutz Favorites data

Implementation Limitations

Dependence on the Skroutz DOM structure, rule-based product categorization, and no continuous price monitoring

Future Work

Evaluation with real users, more resilient data extraction, and support for additional online retailers

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Thank you

Questions?

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